

# Final Project

Austin Wisness  
University of Missouri - St. Louis

May 2024

# Contents

|          |                                         |           |
|----------|-----------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                     | <b>3</b>  |
| <b>2</b> | <b>Data</b>                             | <b>3</b>  |
| 2.1      | Dataset visuals . . . . .               | 3         |
| <b>3</b> | <b>Phase 1</b>                          | <b>4</b>  |
| 3.1      | Task One: Overfit Model . . . . .       | 4         |
| 3.1.1    | First Model . . . . .                   | 4         |
| 3.1.2    | Results . . . . .                       | 5         |
| 3.1.3    | Second Model . . . . .                  | 6         |
| 3.1.4    | Results . . . . .                       | 6         |
| 3.2      | Task Two: Basic CNN . . . . .           | 7         |
| 3.2.1    | Third Model . . . . .                   | 7         |
| 3.2.2    | Results . . . . .                       | 8         |
| 3.2.3    | Fourth Model . . . . .                  | 9         |
| 3.2.4    | Results . . . . .                       | 10        |
| 3.3      | Task Three: Regularization . . . . .    | 10        |
| 3.3.1    | Fifth Model . . . . .                   | 10        |
| 3.3.2    | Results . . . . .                       | 11        |
| 3.3.3    | Sixth Model . . . . .                   | 12        |
| 3.3.4    | Results . . . . .                       | 13        |
| <b>4</b> | <b>Phase 2</b>                          | <b>14</b> |
| 4.1      | Task Four: Pre-trained models . . . . . | 14        |
| 4.1.1    | Seventh Model . . . . .                 | 14        |
| 4.1.2    | Results . . . . .                       | 14        |
| 4.2      | Classification model . . . . .          | 15        |
| 4.2.1    | UNet Model . . . . .                    | 15        |
| 4.2.2    | Results . . . . .                       | 15        |
| <b>5</b> | <b>Conclusion</b>                       | <b>16</b> |

# 1 Introduction

The task of identifying different kinds of plants is a very tedious task if you don't have the knowledge on hand. It would take even longer identifying if it's succumbed to disease or not. To help combat this problem there needs to be an AI agent that can identify areas of the plant that could be diseased and what plant it is. This will require two models handling the tasks of identifying the plant with it's disease and identifying areas of the plant that are diseased, if present.

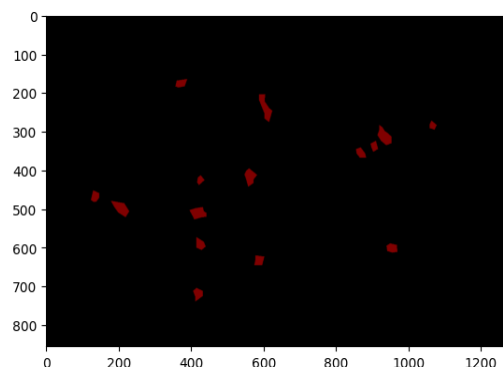
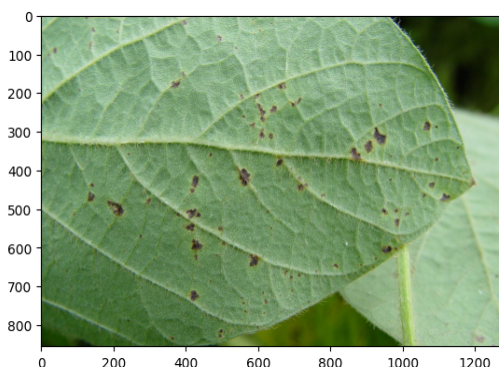
## 2 Data

The data was cured from two sources where one dataset is for the image segmentation with images and corresponding masks. The other is the PlantDoc dataset for the classification model which identify plants and diseases. The plants trained in this model are

- Apple leaf
- Bell Pepper leaf
- Blueberry leaf
- Cherry leaf
- Corn leaf
- Peach leaf
- Potato leaf
- Raspberry leaf
- Soybean leaf
- Squash leaf
- Strawberry leaf
- Tomato leaf
- Grape leaf.

The first dataset is from <https://www.kaggle.com/datasets/fakhrealam9537/leaf-disease-segmentation-dataset/data> while the other dataset is from <https://github.com/pratikkayal/PlantDoc-Dataset>

### 2.1 Dataset visuals



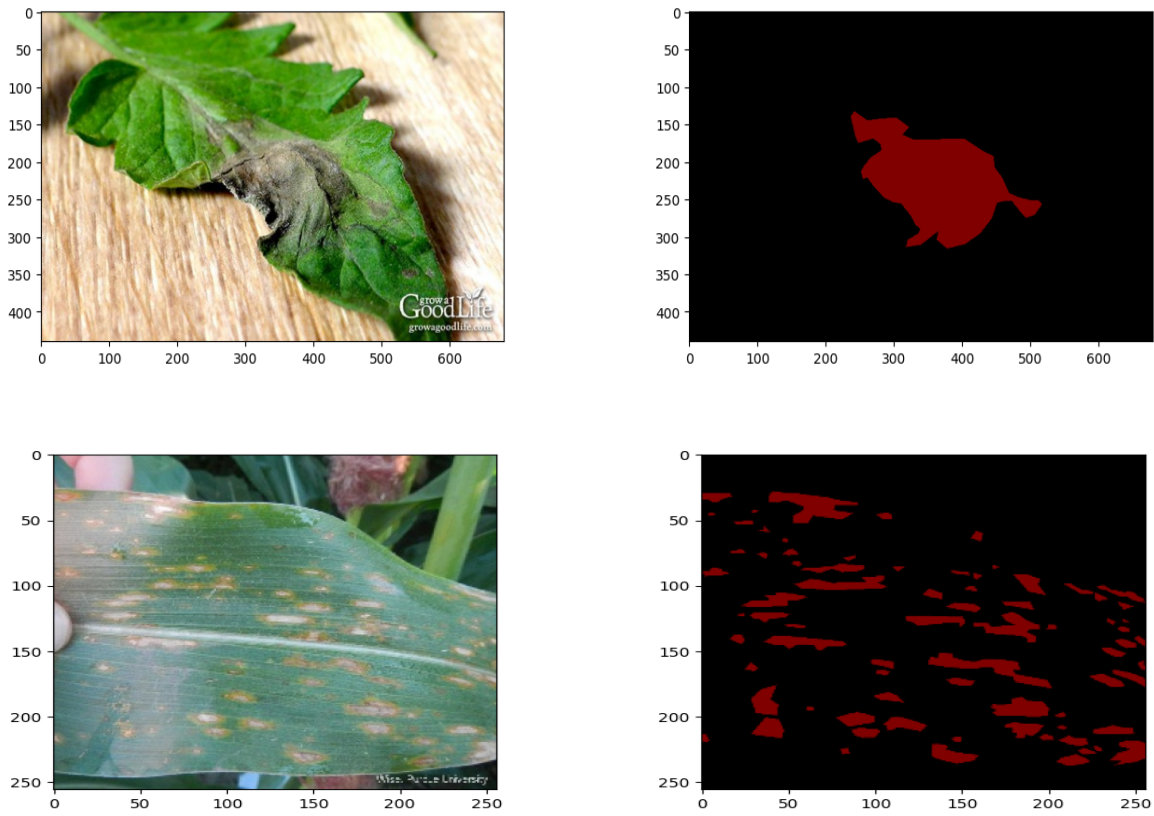


Figure 3: Images and their masks

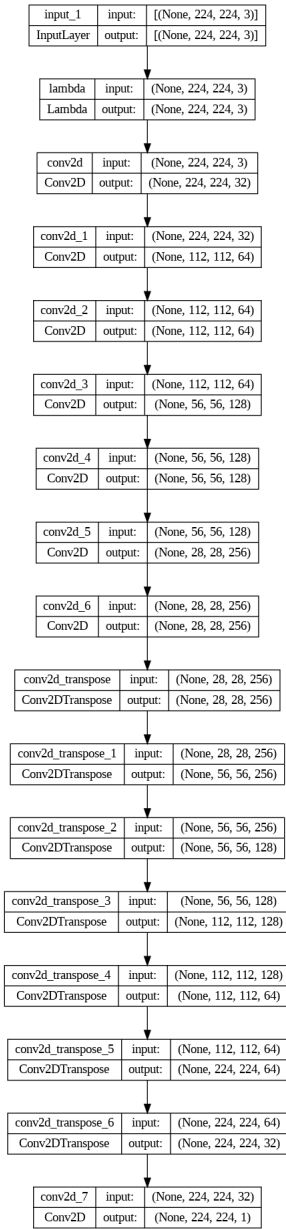
### 3 Phase 1

The first phase is steps in creating a model for the problem. First is overfitting the model to find the capacity of an architecture that works well. After that is building the base convolution neural network to gain better validation accuracy. Following that is to apply regularization techniques to further increase the validation accuracy.

#### 3.1 Task One: Overfit Model

##### 3.1.1 First Model

The first model has an upsampling block and a downsampling block used for image segmentation models. It has 2915265 parameters. This model has the following layout:



### 3.1.2 Results

This model already does fairly well at over 80 percent accuracy. It begins to over fit after 5 epochs. The prediction shows the mask the model created.

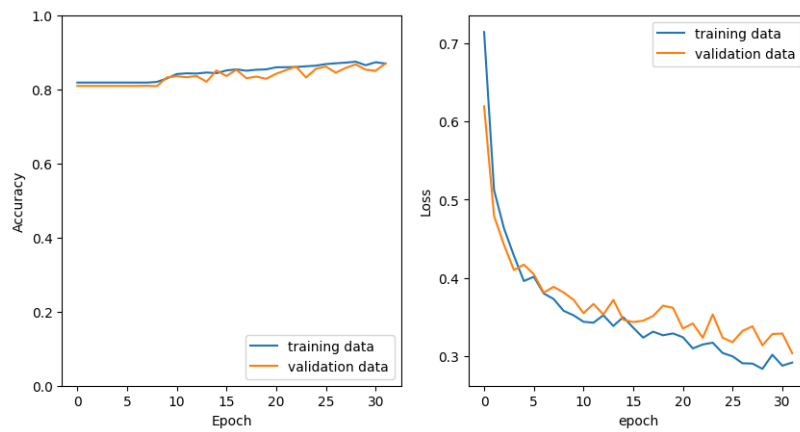


Figure 4: Model 2 results

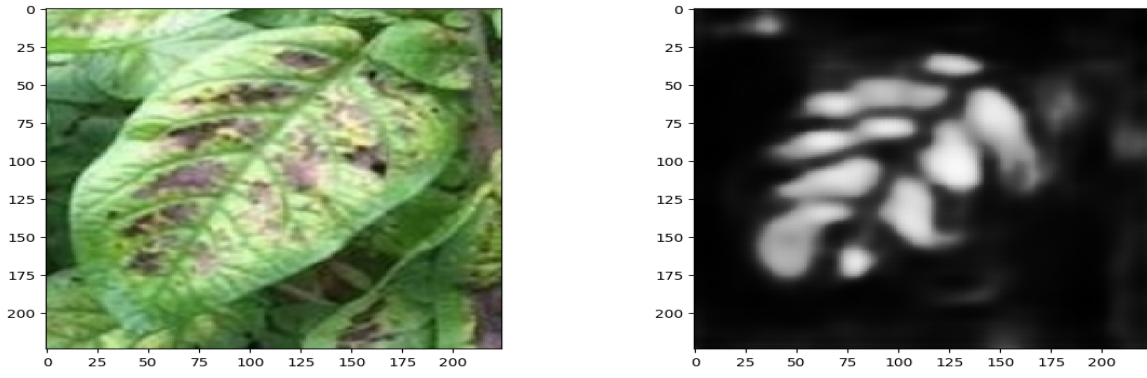
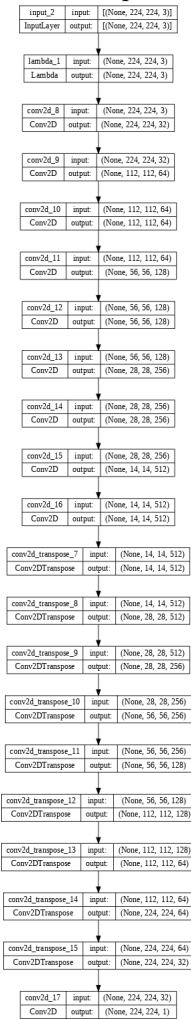


Figure 5: Images and their masks

### 3.1.3 Second Model

The second model has an upsampling block and a downsampling block used for image segmentation models. It has 11764673 parameters. This model has the following layout:



### 3.1.4 Results

This model already shows improvement with over 85 percent accuracy. It begins to over fit around 20 epochs. The prediction shows the mask the model created.

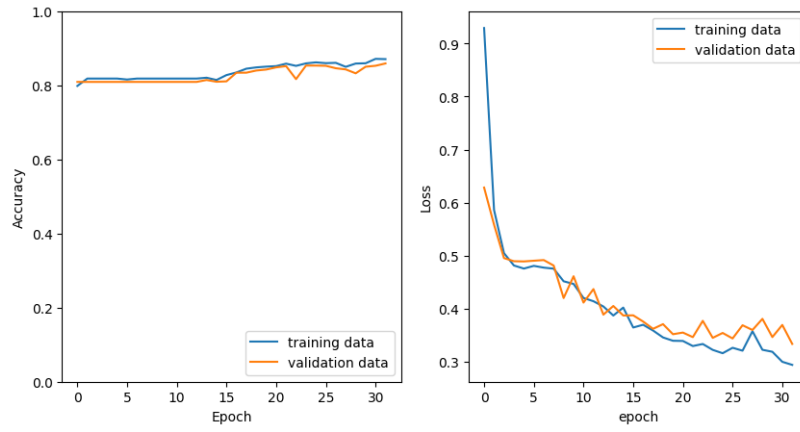


Figure 6: Model 2 results

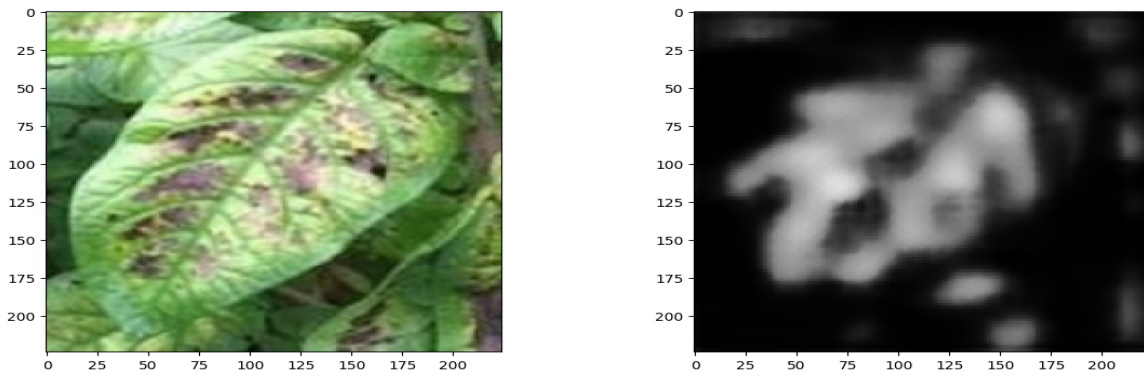
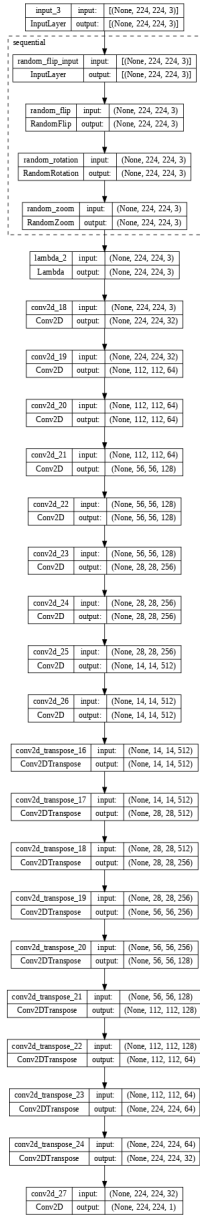


Figure 7: Image and predicted mask

## 3.2 Task Two: Basic CNN

### 3.2.1 Third Model

The third model has an upsampling block and a downsampling block used for image segmentation models. Along with a data augmentation layer to rotate, flip and zoom the images. It has 11764417 parameters. This model has the following layout:



### 3.2.2 Results

This model had some changes like data augmentation layer and changes to the last output layer. This one starts to over fit after 30 epochs but struggles to learn. The accuracy did not improve at all.

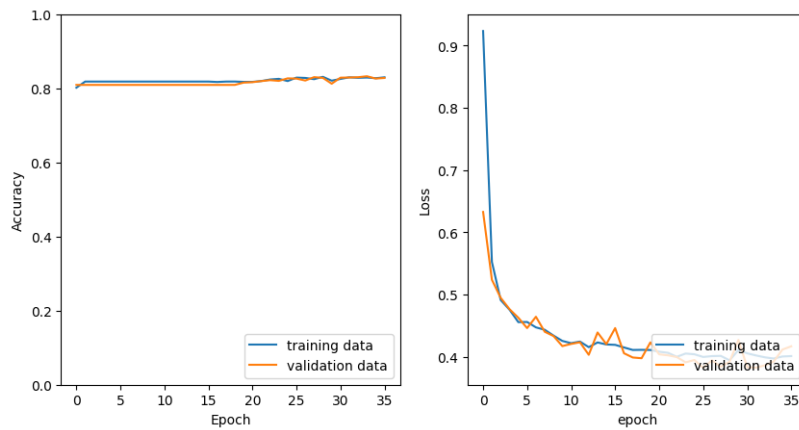


Figure 8: Model 3 results



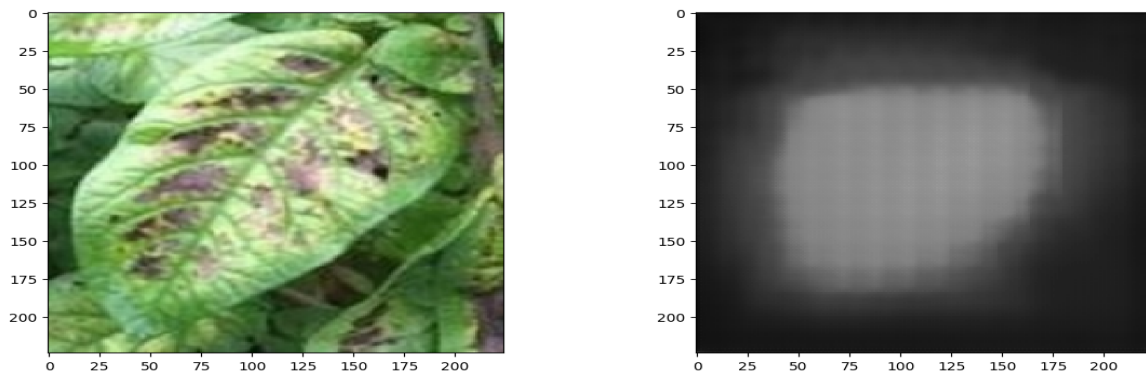
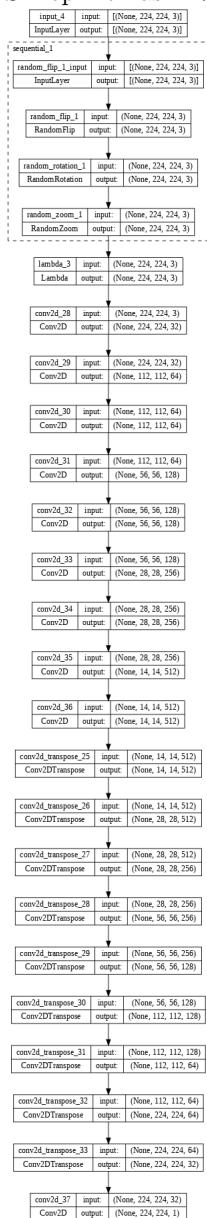


Figure 9: Image and predicted mask

### 3.2.3 Fourth Model

The fourth model has an upsampling block and a downsampling block used for image segmentation models. Along with a data augmentation layer to rotate, flip and zoom the images. This model uses a different optimizer, RMSProp. It has 11764417 parameters. This model has the following layout:



### 3.2.4 Results

This model had the optimizer changes to rmsprop instead of adam. This model did not improve. It began to over fit around 10 epochs. It did not learn at all with no improvement in accuracy.

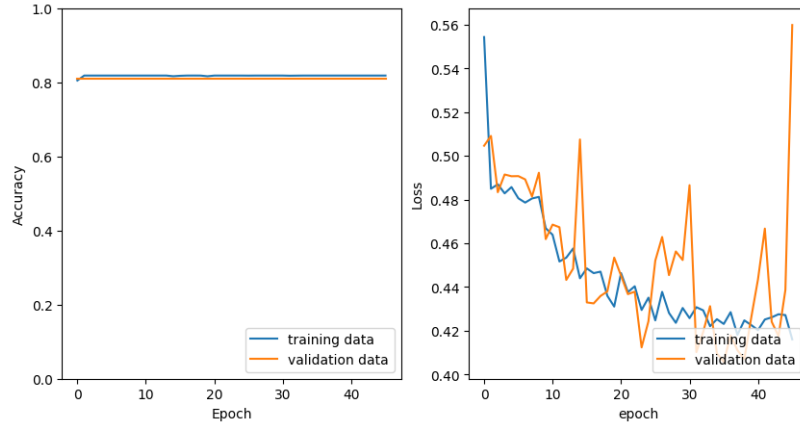


Figure 10: Model 4 results

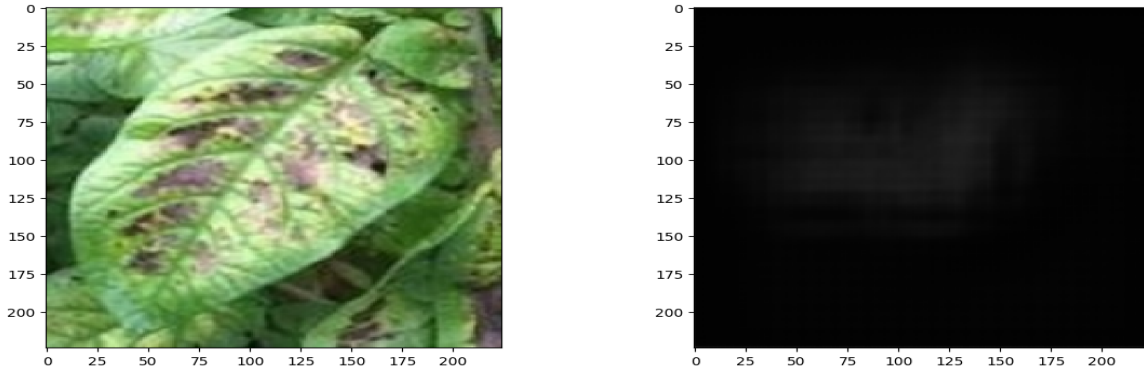
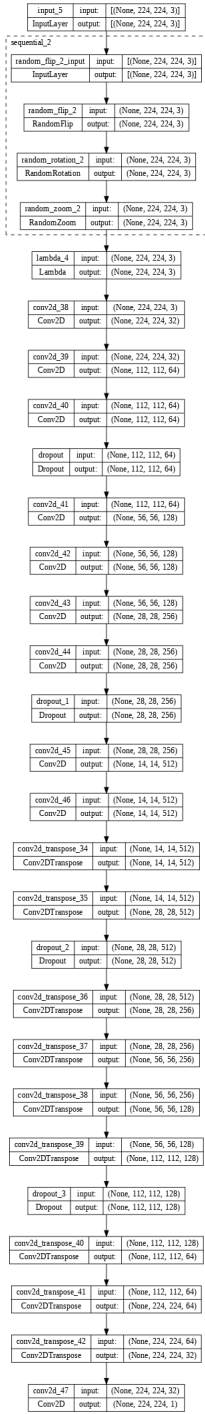


Figure 11: Image and predicted mask

## 3.3 Task Three: Regularization

### 3.3.1 Fifth Model

The fifth model has an upsampling block and a downsampling block used for image segmentation models. These blocks also have two dropout layers along with a data augmentation layer to rotate, flip and zoom the images. It has 11764417 parameters. This model has the following layout:



### 3.3.2 Results

This model had some dropout layers added and using the adam optimizer again. This model had some improvement. It began to over fit around 30 epochs and managed to train over 40 epochs. Which is a big improvement.

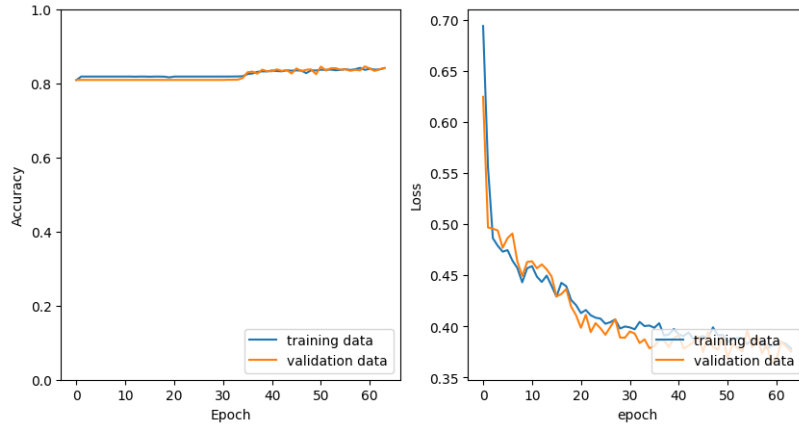


Figure 12: Model 5 results

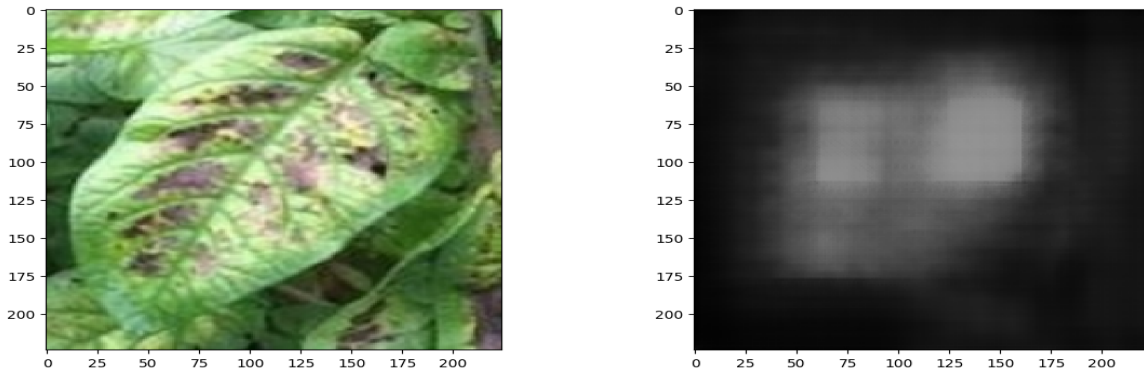
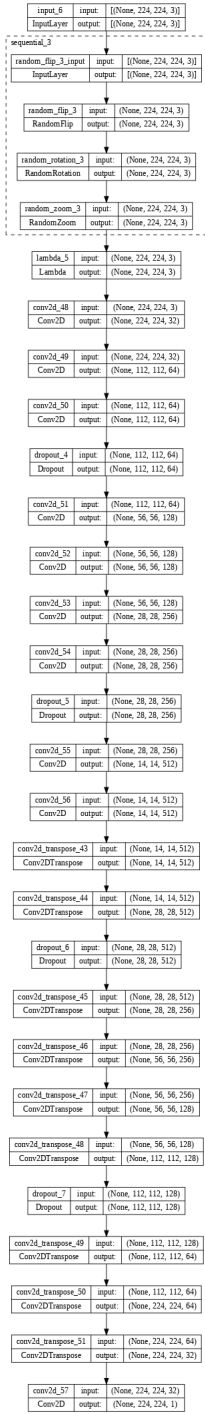


Figure 13: Image and predicted mask

### 3.3.3 Sixth Model

The fifth model has an upsampling block and a downsampling block used for image segmentation models. These blocks also have two dropout layers along with a data augmentation layer to rotate, flip and zoom the images. The first layer of the downsampling block and the last layer of the upsampling block have L1 regularization applied. It has 11764417 parameters. This model has the following layout:



### 3.3.4 Results

This model had l1 regularization in beginning and end layers. It drastically reduced overfitting but the model did not learn well.

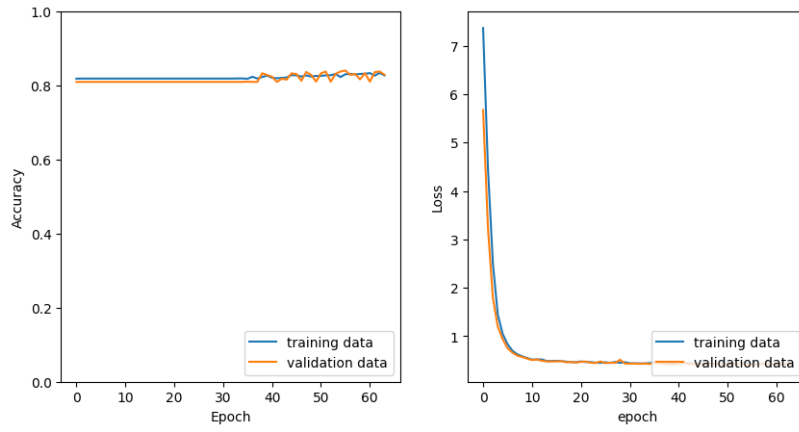


Figure 14: Model 6 results

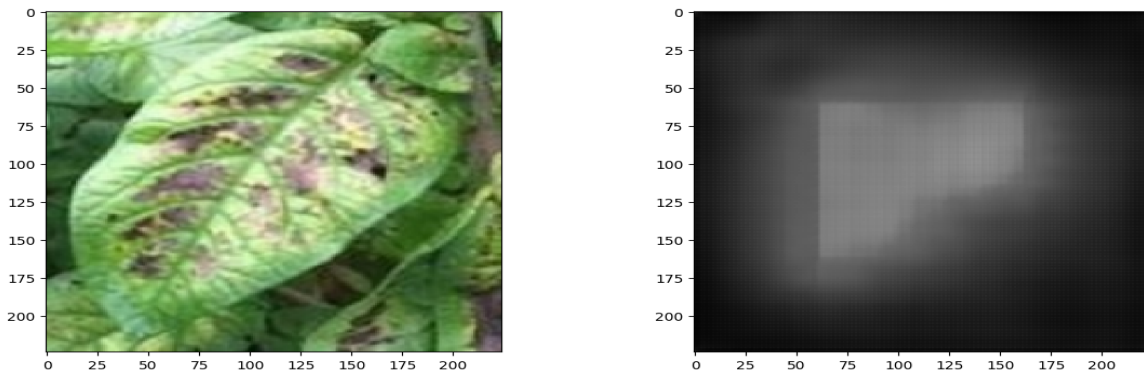


Figure 15: Image and predicted mask

## 4 Phase 2

### 4.1 Task Four: Pre-trained models

#### 4.1.1 Seventh Model

This model has too many layers to provide a model diagram since it uses the mobilenetv2 model as the down-sample block. It also uses pix2pix for the upsample layers block.

#### 4.1.2 Results

The results were not as expected but the mask it predicted looks very close to getting the right area shaded. It is hard to see in the mask but you can see the pattern of the leaf. It had a interesting spike after 30 epochs.

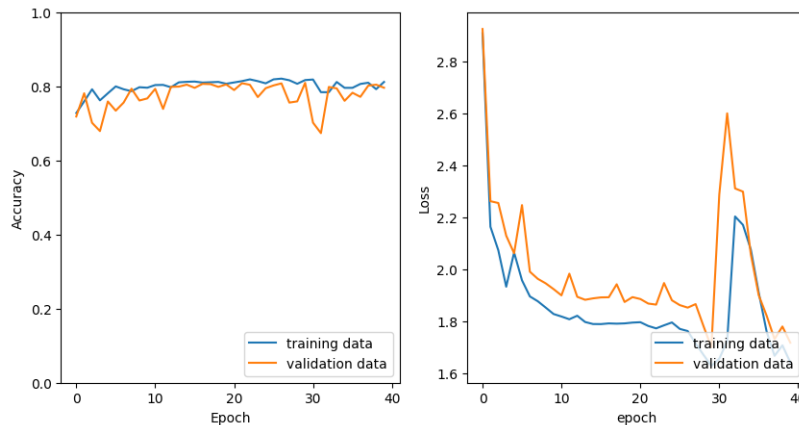


Figure 16: Model 7 results

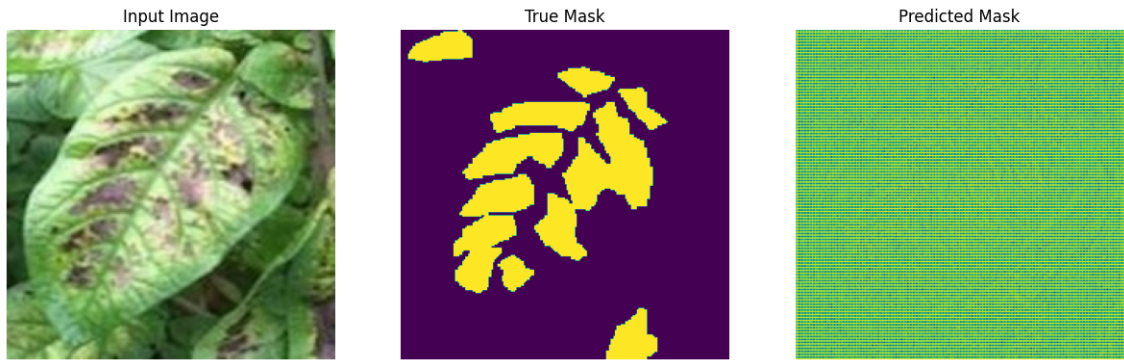
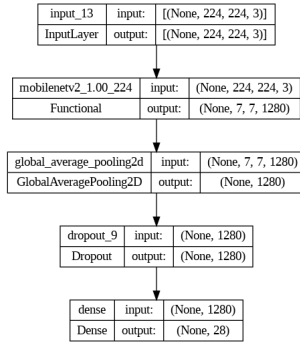


Figure 17: Model 7 results

## 4.2 Classification model

### 4.2.1 UNet Model

The classification model uses a UNet architecture with the mobilenetv2 pre-trained model as the base. It has 2293852 parameters.



### 4.2.2 Results

This model got great results with 96.13 percent accuracy.

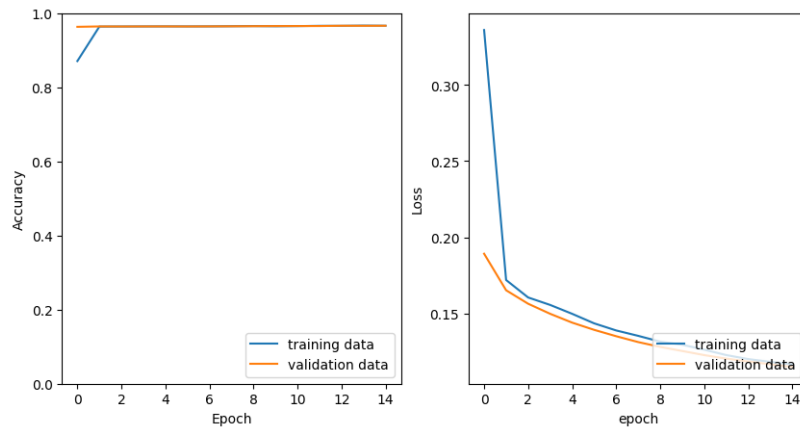


Figure 18: Model 8 results

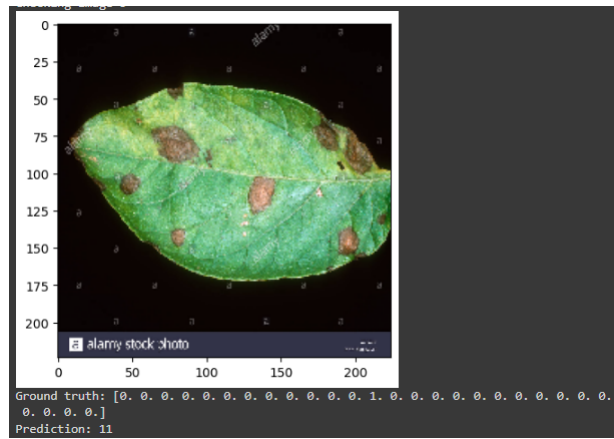


Figure 19: Model 8 prediction

## 5 Conclusion

The end results were better than expected. The best image segmentation model was surprising the second model with a high of 85 percent validation accuracy in training. The classification model turned out great with a 96.13 percent accuracy on a test set. There are more parameters that could be changed to improve accuracy. The image segmentation dataset could also use better mask quality to help the model train on. Manual annotation is a time consuming process, so it is hard to find quality image segmentation datasets in the public domain.